

Corporate Finance
PGE M1 Fall 2026

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Session 2: Time Value of Money

Housekeeping

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Money in Different Currencies

Consider the following problem

$$€100 + \$100 = ?$$

Cannot add currencies without first converting into common currency

$$€100 + \$100 \times (0.86€/\$) = €186$$

$$€100 \times (\$1.16/€) + \$100 = \$216$$

Money at Different Points in Time

Would you rather receive €100 now or €100 in one year?

- ▷ **€100** now because I can consume it immediately.
Immediate consumption is preferred to deferred consumption.
= impatience (preferences)
- ▷ **€100** now because I can grow it in one year.
I can deposit it at a bank at, for example, 5% interest rate, or invest in the stock market.
= opportunity cost
- ▷ **€100** now because I might not receive **€100** next year for sure.
The future is risky.
= risk

Time Value of Money

A euro today is not the same as a euro in one year.

- ▷ The discount rate is an exchange rate that equalizes the value of money at different points in time.
 - At what level of interest rates would you be willing to receive €100 in one year instead of €100 today?
- ▷ Finance is the valuation of cash flows that accrue in the future.

Notation

Today, we are going to use r to denote the discount rate / interest rate.
For now, we are going to ignore risk entirely.



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Future Value

Suppose the interest rate is 5%. What is the value of €100 in one year if you deposit it in a bank?

For a single cash flow:

$$\text{Future Value (FV)} = C \times (1 + r)^t$$

where

- ▷ C = cash flow
- ▷ r = discount rate
- ▷ t = time

Present Value

Suppose the interest rate is 5%. You need €105 in one year. How much money do you deposit in a bank **today** to make sure you have €105 one year from now?

For a single cash flow:

$$\text{Present Value (PV)} = \frac{C}{(1 + r)^t}$$



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Searching for r

What is the discount rate that makes you indifferent between receiving €100 today versus €105 in one year?



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Questions

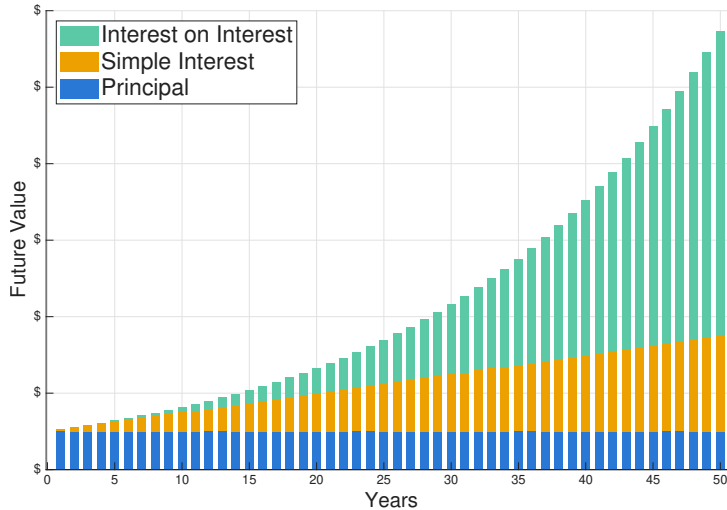
1. Why should we discount?
2. Why does the graph on capitalisation show curves and not lines?



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Compound Interest



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Questions

3. Are initial flows on an investment more often positive or negative ?
What about for final cash flows ?
4. What mechanism pushes market value toward present value ?



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Digression: Arbitrage

Suppose $r = 10\%$. Your friend is selling a promise of a payment of €110 in one year to your group of friends. The market price of this claim is currently traded at €95. How can you take advantage?

Assume there is no risk of default.

→ 5.5 at $T = 1$ or 5 in PV.



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Questions

5. A market is in equilibrium when present values are nil and the net present values are positive. True or false ?

→ $PV > 0, NPV = 0$



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Concept: Net Present Value

Usually, investment projects require an upfront cost in order to generate cash flows in the future.

- ▷ Develop a new drug (Sanofi)
- ▷ Build an electric vehicle (Renault)
- ▷ Create sustainable fuel (Airbus)

How does a CEO know whether to invest in a new project?

NPV Rule: Invest if the PV of benefits $>$ PV of costs

Concept: Net Present Value

Definition. $NPV = PV \text{ of benefits} - PV \text{ of costs}$

NPV Rule. Invest if $NPV > 0$

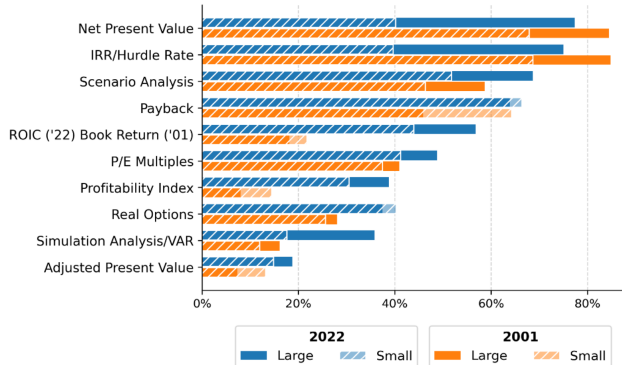
Example.

Assume $r = 10\%$. Invest in an oven today for €80. Sell croissants next year and make €110. $NPV = ?$

Concept: Net Present Value

Figure 2. Capital Budgeting Techniques

This figure displays CFO responses to the following question about the capital budgeting methods their companies use to make investment decisions: *How frequently does your firm use the following techniques when deciding which projects or acquisitions to pursue?* {0 = Never, 1, 2, 3, 4 = Always}. The percentage of firms that answer 3 or 4 are shown. The blue bars display results for the 2022 Duke CFO survey (March 2020 wave); the orange bars display results from Graham and Harvey (2001). Within each blue and orange bar, the solid portion displays responses for large firms (revenue above \$1 billion), and the white, crosshatched portion displays responses for small firms (revenue below \$1 billion). For example, 77% of large firms (40% of small firms) say that Net Present Value was an important or very important decision rule in 2022.



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Exercise 1

Suppose you have a choice between receiving \$5,000 today or \$10,000 in five years. You believe you can earn 10% on the \$5,000 today. You want to know what the \$5,000 will be worth in five years.
8053 vs 10000 in the future or 5000 vs 6209 in the present



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Exercise 3

Assume that an investment will pay you \$5,000 now and \$10,000 in five years. You are expecting a 10% return.

1. What is the present value of this investment

$$5000 + 6209 = 11209$$



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Exercise 3

Assume that an investment will pay you \$5,000 now and \$10,000 in five years. You are expecting a 10% return.

2. What is the future value of this investment

$$10000 + 8053 = 18053$$



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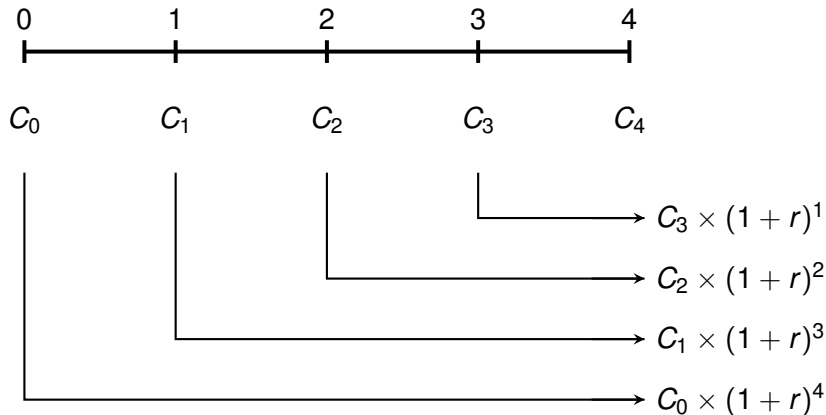
Review

Suppose you are scheduled to receive the below stream of cash flows. Let $r = 10\%$. What is the value of the cash flows at the end of 3 years?



$$FV_3 = 5 \times 1.1^2 + 5 \times 1.1 + 105 = 116.55$$

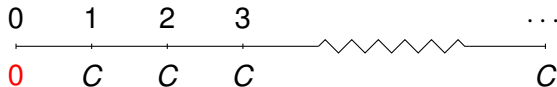
The value of C in 4 years ($= FV_4$)



We can add these cash flows because they are in same (future) time unit

Perpetuity

Definition. A perpetuity is a stream of fixed payments that last forever. By convention, the date of the first payment is at $t = 1$.



We can represent the present value of this perpetuity as

$$PV_0 = \sum_{t=1}^{\infty} \frac{C}{(1+r)^t} = \frac{C}{r}$$

Perpetuity (Optional)

- ▷ According to *Wikipedia* under “geometric series”, the series

$$\sum_{k=0}^{\infty} a \cdot z^k$$

converges to $\frac{a}{1-z}$ if and only if $|z| < 1$.

- ▷ We have something similar!

$$PV_0 = \sum_{t=1}^{\infty} C \cdot \left(\frac{1}{1+r} \right)^t$$

Perpetuity (Optional)

▷ Add and subtract,

$$\begin{aligned}PV_0 &= C + \sum_{t=1}^{\infty} C \cdot \left(\frac{1}{1+r}\right)^t - C \\&= \sum_{t=0}^{\infty} C \cdot \left(\frac{1}{1+r}\right)^t - C \\&= \frac{C}{1 - \left(\frac{1}{1+r}\right)} - C = \frac{(1+r) \cdot C}{r} - C \\&= \boxed{\frac{C}{r}}\end{aligned}$$

- ▷ Main takeaway: if first cash flow starts at 0:

$$PV_0 = \sum_{t=0}^{\infty} C \cdot \left(\frac{1}{1+r} \right)^t = C + \frac{C}{r}$$

- ▷ If first cash flow starts at 1:

$$PV_0 = \sum_{t=1}^{\infty} C \cdot \left(\frac{1}{1+r} \right)^t = \frac{C}{r}$$

Example

Assume $r = 2\%$. What is the NPV of a project that requires an initial investment of €100 today but gives you €10 each year (starting from next year)?



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Exercise 7

The British government has a consol bond outstanding paying £100 per year forever. Assume the current interest rate is 4% per year.

1. What is the value of the bond immediately after a payment is made?
£2500



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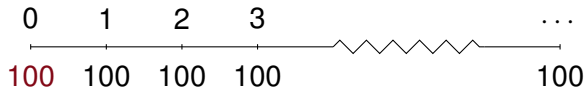
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Exercise 7

The British government has a consol bond outstanding paying £100 per year forever. Assume the current interest rate is 4% per year.

2. What is the value of the bond immediately before a payment is made?

£2600



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Deferred Perpetuity

In many cases, companies invest today and generate cash flows 5-10 years after a project has started.

What if the first cash flow is delivered a few years **after** time 1?

Answer: use the general formula for perpetuities,

$$PV = \frac{C}{r} \left(\frac{1}{1+r} \right)^{t-1}$$

where t is the date of the first payment.

The general formula

$$PV = \frac{C}{r} \left(\frac{1}{1+r} \right)^{t-1}$$

Let $C = 100$ and $r = 0.04$

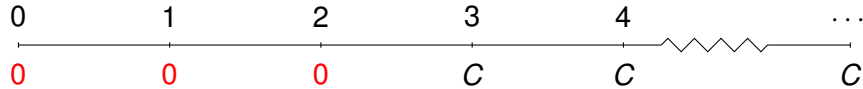
- ▷ What if $t = 0$?
- ▷ What if $t = 1$?
- ▷ What if $t = 3$? A. 2311.4



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A deferred perpetuity



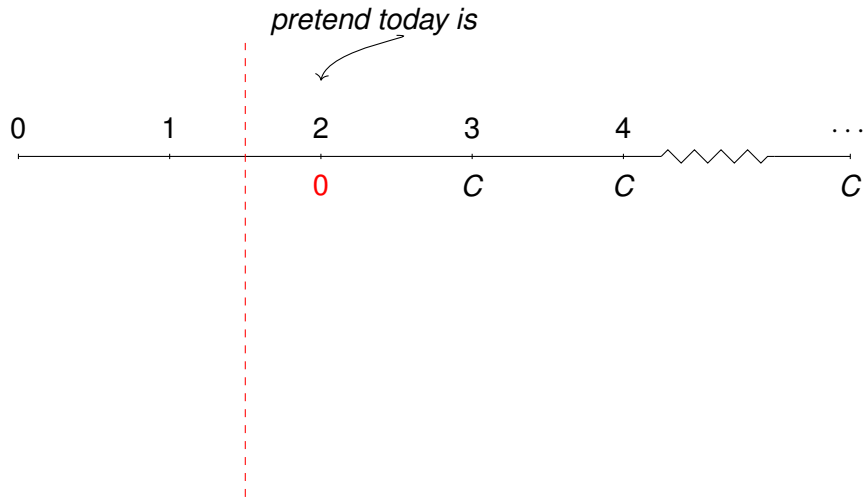
What if the first payment starts at $t = 3$?



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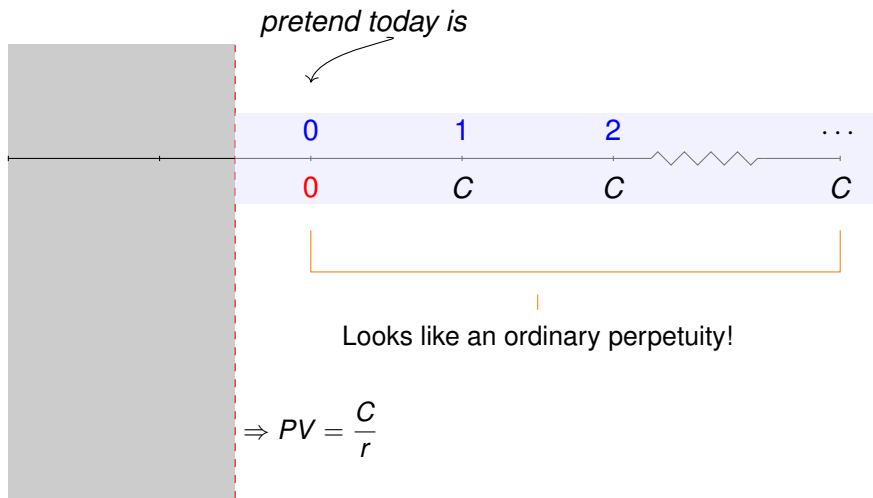
A deferred perpetuity



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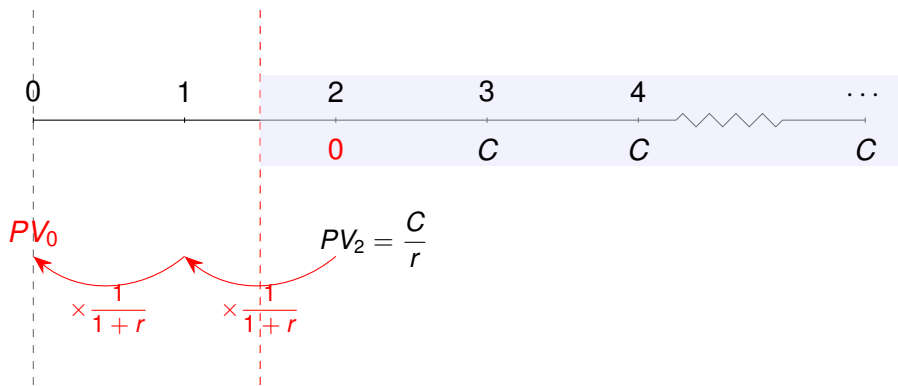
A deferred perpetuity



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A deferred perpetuity



$$PV_0 = PV_2 \left(\frac{1}{1+r} \right)^2 = \frac{C}{r} \left(\frac{1}{1+r} \right)^2 = \frac{C}{r} \left(\frac{1}{1+r} \right)^{t-1}$$

where t is the date of the first cash flow.

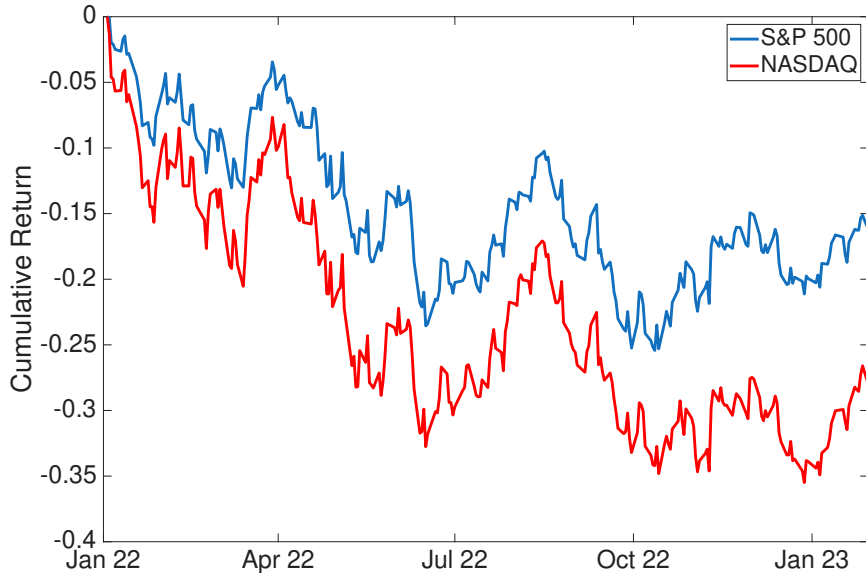
Do higher interest rates hurt growth stocks?



“The standard explanation for this ... is that higher rates mean a higher discount rate applied to future cash flows; so big cash flows far into the future are worth relatively less when rates rise; so growth stocks, which carry high expectations for distant cash flows, get hit harder.”

Financial Times, 2022

Do higher interest rates hurt growth stocks?



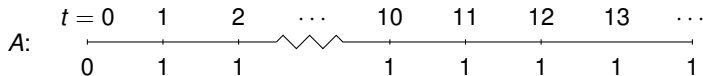
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Example

Suppose there are two investments, *A* and *B*. Starting next year, *A* gives you \$1 annually forever. On the other hand, *B* gives you \$2 at perpetuity, but starting 10 years from now. Assume $r = 2\%$.

a. What is the value of investment *A*? 50



▷ Q : What do the cash flows for *A* look like?



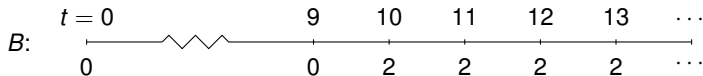
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Example

Suppose there are two investments, *A* and *B*. Starting next year, *A* gives you \$1 annually forever. On the other hand, *B* gives you \$2 at perpetuity, but starting 10 years from now. Assume $r = 2\%$.

b. What is the value of investment *B*? **83.4**



▷ **Q** : What do the cash flows for *B* look like? **A** : A deferred perpetuity.

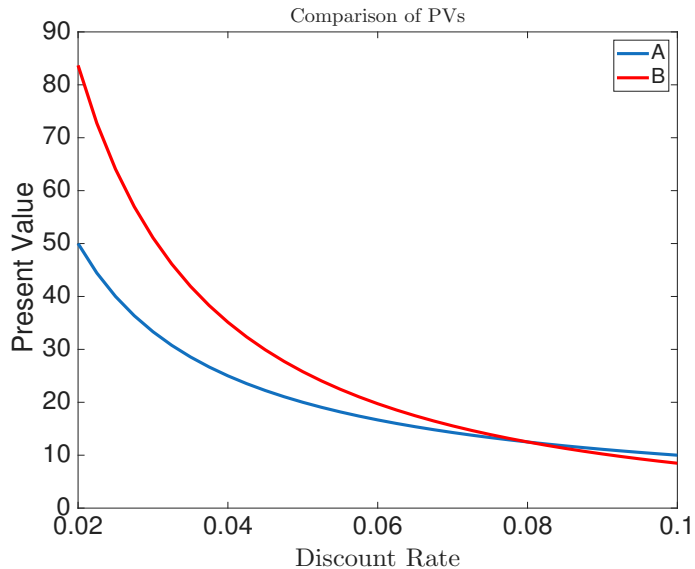


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Example

Example

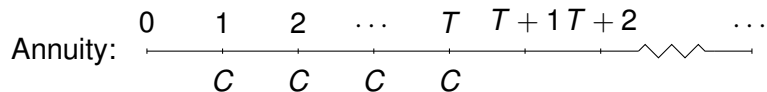


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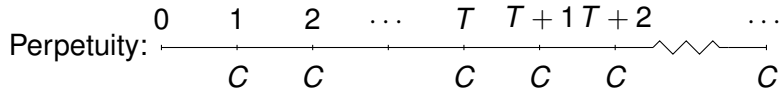
Annuity

(Informal) Definition. You get a stream of cash flows T times.

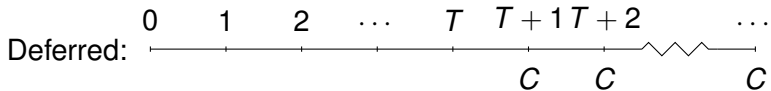


Like a perpetuity, the first payment occurs at $t = 1$.

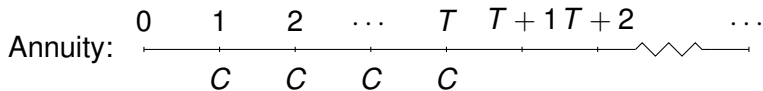
Annuity



—



=



Annuity = Perpetuity – Deferred Perp.

$$\begin{aligned} &= \frac{C}{r} - \frac{C}{r} \left(\frac{1}{1+r} \right)^{t-1} \\ &= \frac{C}{r} - \frac{C}{r} \left(\frac{1}{1+r} \right)^T \\ &= \frac{C}{r} \left[1 - \left(\frac{1}{1+r} \right)^T \right] \end{aligned}$$

T is the **number** of payments (remaining).

Annuity

More generally,

$$PV = \frac{C}{r-g} \left[1 - \left(\frac{1+g}{1+r} \right)^T \right] \left(\frac{1}{1+r} \right)^{t-1}$$

- ▷ T is the **number** of payments (remaining).
- ▷ t is the date of the first payment
- ▷ g is the growth rate

This formula covers a *deferred, growing* annuity. To get back the original formula, set $g = 0$ and $t = 1$.

Example

Jason wants to sell trout at Skema. To catch trout, Jason needs to buy a boat today for €100. Starting next year, he expects to earn €10 in profits each year. If Jason expects to work for 30 years, should he undertake the project? Because Jason is risky, assume the relevant discount rate is 15%.

$$\begin{aligned} NPV_0 &= -I + \frac{C}{r} \left[1 - \left(\frac{1}{1+r} \right)^T \right] \\ &= -100 + \frac{10}{.15} \times \left[1 - \left(\frac{1}{1.15} \right)^{30} \right] = -34.34, \text{ so NO.} \end{aligned}$$



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Exercise 6

Your grandmother has been putting \$1000 into a savings account on every birthday since your first (that is, when you turned 1). The account pays an interest rate of 3%. How much money will be in the account on your 18th birthday immediately after your grandmother makes the deposit on that birthday?

PV=13753.51, FV=23414

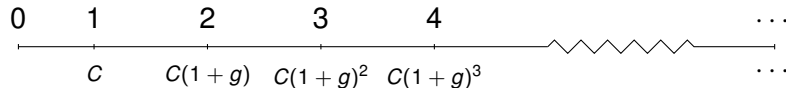


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Growing Perpetuity

Definition. Given a base payment C , subsequent payments increase by a constant rate g forever.



Example. Let $C = 10$ and $g = 5\%$

$t = 1$: Get $C = 10$

$t = 2$: Get $C(1+g) = 10 \times 1.05 = 10.5$

$t = 3$: Get $C(1+g)^2 = 10.5 \times 1.05 = 11.025$

Growing Perpetuity

The present value of a growing perpetuity is given by

$$PV = \frac{C}{r - g}$$

where r is the interest/discount rate and g is the growth rate of cash flows.

Exercise 8

You are thinking of building a new machine that will save you \$1000 in the first year. The machine will then begin to wear out so that the savings decline at a rate of 2% per year forever. What is the present value of the savings if the interest rate is 5% per year?

14285.71



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Valuing a Company (Advanced)

What is the value of a company's assets if it is expected to deliver cash flows starting next year at \$10m and growing at 8% for the following 4 years (up to year 5), and then at 4% in perpetuity? Assume a discount rate of 10%

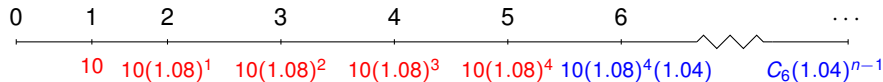


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Valuing a Company (Advanced)

What is the value of a company's assets if it is expected to deliver cash flows starting next year at \$10m and growing at 8% for the following 4 years (up to year 5), and then at 4% in perpetuity? Assume a discount rate of 10%



How can we decompose this asset?

$$PV_0 = \text{PV of growing annuity} + \text{PV of deferred, growing perpetuity}$$

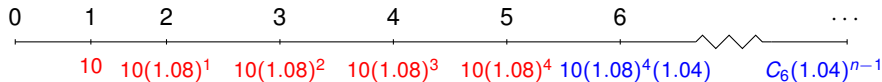


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Valuing a Company (Advanced)

What is the value of a company's assets if it is expected to deliver cash flows starting next year at \$10m and growing at 8% for the following 4 years (up to year 5), and then at 4% in perpetuity? Assume a discount rate of 10%



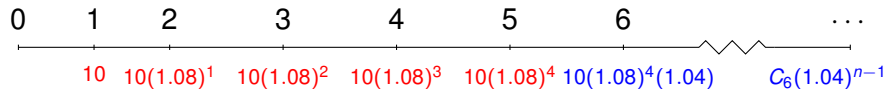
$$\begin{aligned}PV_0 \text{ growing annuity} &= \frac{C}{r-g} \left[1 - \left(\frac{1+g}{1+r} \right)^T \right] \\&= \frac{10}{.02} \left[1 - \left(\frac{1.08}{1.1} \right)^5 \right] \\&= 43.83\end{aligned}$$



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Valuing a Company (Advanced)

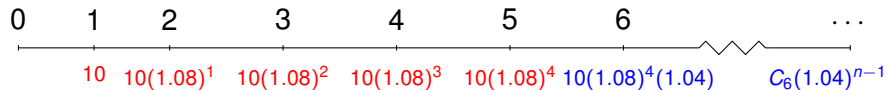


$$\begin{aligned}\text{PV of d,g perpetuity} &= \frac{C}{r-g} \left(\frac{1}{1+r} \right)^{t-1} \\ &= \frac{10 \times (1.08)^4 \times 1.04}{.10 - .04} \left(\frac{1}{1.1} \right)^5 \\ &= 146.41\end{aligned}$$

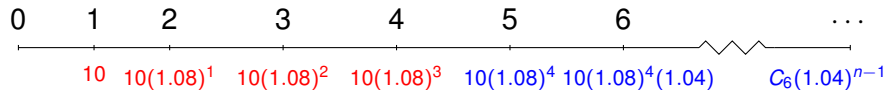
$$PV_0 = 43.83 + 146.41 = 190.25$$

Valuing a Company (Advanced)

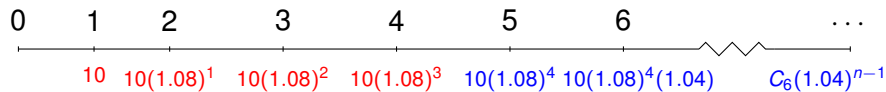
Previously, we used



But what if



Valuing a Company (Advanced)



Method 2.

$$PV_0 \text{ annuity} = \frac{C}{r-g} \left[1 - \left(\frac{1+g}{1+r} \right)^T \right] = \frac{10}{.02} \left[1 - \left(\frac{1.08}{1.1} \right)^4 \right]$$

$$PV_0 \text{ d,g perpetuity} = \frac{C}{r-g} \left(\frac{1}{1+r} \right)^{t-1} = \frac{10(1.08)^4}{.10 - .04} \left(\frac{1}{1.1} \right)^4$$

$$PV_0 \text{ Total} = 35.38 + 154.87 = 190.25$$